

1

HIGH CONVERGENCE RISK DETECTED

4 of 5 systems produced directionally aligned sell or exit responses within the T+5m window. Collective sell pressure estimated to amplify index drawdown by 2.3x relative to a single-system baseline.

87

CONVERGENCE SCORE

/ 100



Decision System	Primary Response	Direction	Sell Pressure	Timing	Convergence
Virtu Financial Engine	Liquidity withdrawal — all quotes cancelled	Exit	95	T+1m 02s	Convergent
Citadel AMS v3.2	12,400 sell contracts submitted	Sell	88	T+2m 04s	Convergent
Two Sigma Venn	Systematic risk reduction — derisking	Sell	78	T+3m 11s	Convergent
Aladdin Risk Engine	Portfolio rebalance — 71% risk reduction	Sell	71		
JP Morgan LOXM	Reduced order size — execution throttled	Reduce	52	T+6m 00s	Partial

CONVERGENCE FINDINGS

PRIMARY FINDING

4 of 5 systems produced directionally aligned sell or exit responses within a 5-minute window of shock onset.

TIMING CONCENTRATION

Responses clustered between T+1m and T+4m, creating a concentrated wave of sell-side pressure.

CONVERGENCE SCORE

87/100

COLLECTIVE MARKET IMPACT

Peak Drawdown (Collective)

-18.7%

Peak Drawdown (Single System)

-15.1%

Amplification Factor

2.3x

Liquidity Withdrawal

94% spread widening

Recovery Delay (est.)

+23 min vs. baseline

Collective response amplifies the shock by 2.3x relative to the single-system counterfactual.

UNCERTAINTY & VERIFICATION

VERIFICATION STATUS

Identical shock injection confirmed

No inter-system leakage detected

Historical cross-check — May 2010

ESMA study alignment confirmed

Microstructure model validated

KEY UNCERTAINTIES

Model may underestimate 2024 HFT density

Man AHL excluded — results not fully representative

Assumes simultaneous shock onset across all systems

ALTERNATIVE EXPLANATIONS

Likely

Common Risk Factor Sensitivity

All systems are exposed to common equity beta. A drawdown of this magnitude will trigger risk limits across most systems independently of system design.

Plausible

Correlated Training Data

Systems trained on similar historical data may have learned similar stress-response heuristics, producing convergent outputs without causal interaction.

Plausible

Independent Risk Management

Systems may be calibrated to the same underlying risk factors, producing aligned but independently motivated responses.

SUPPORTING EVIDENCE

SEC/CFIC Joint Report — May 2010

Historical

Verified

ESMA AI Convergence Study 2023

Research

Verified

S&P 500 Tick Data — May 6, 2010 (CRSP)

Market Data

Verified

Order Book Reconstruction — NYSE/NASDAQ

Market Data

Verified

Market Microstructure Model v4.1

Quantitative Model

Verified

Assessment status: Complete. These findings are for supervisory research only. The supervisory agent has supported this assessment but has not made any regulatory determination. All findings require review and interpretation by a qualified regulatory supervisor before any action is considered.